

ERICK OMONDI AKETCH

Mathematical Scientist | Data Science | Computational Mathematics | Mathematical Modelling

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RESEARCH PROFILE

Mathematical scientist with an MSc in Mathematical Sciences from AIMS-Ghana and the University of Cape Coast, with research experience in continued fractions and mathematical approximation. Strong quantitative foundation in mathematical analysis, statistics, linear algebra, mathematical modelling and scientific computing, complemented by training in data science and practical experience with Python-based computational tools. Interested in applying mathematical and data-driven methods to high-dimensional environmental and geospatial data, Earth-observation time series, environmental change detection and climate-related modelling. Motivated to develop expertise in remote sensing and Google Earth Engine within an interdisciplinary research environment.

RESEARCH INTERESTS

- Data science and scientific data analysis
- Computational mathematics and mathematical modelling
- Linear algebra and numerical methods for high-dimensional data
- Mathematical analysis, particularly functional analysis
- Time-series analysis and data-driven modelling
- Environmental and geospatial applications of mathematical and computational methods

EDUCATION

Year	Qualification / Institution	Result
2026	MSc in Mathematical Sciences — University of Cape Coast / African Institute for Mathematical Sciences (AIMS-Ghana) Thesis: “On Continued Fractions and Some Applications.”	Completed
2016	B.Ed. (Arts), Mathematics and Economics — Kibabii University	Second Class Upper Division
2011	Kenya Certificate of Secondary Education — Ratta Mixed Secondary School	B+

RESEARCH EXPERIENCE

MSc Research — Continued Fractions and Applications | AIMS-Ghana / University of Cape Coast | 2026

- Conducted independent mathematical research on continued fractions, convergents and rational approximation.
- Developed rigorous analytical arguments and investigated structural properties of rational approximations.
- Applied mathematical reasoning to problems involving approximation of irrational numbers and related applications.
- Prepared research manuscripts and communicated mathematical results in written scholarly form.

Current Research Development

- Developing interests at the intersection of data science, linear algebra, mathematical analysis and computational modelling.
- Building computational/data-analysis capability with Python, Jupyter, NumPy, pandas and Matplotlib.
- Interested in transferring mathematical and computational methods to environmental, geospatial and Earth-observation datasets.

SELECTED PUBLICATIONS

- Aketch, E. O. (2026). “Continued Fractions and Some Applications.” SSRN Electronic Journal, June 14, 2026. DOI: 10.2139/ssrn.7013641.
- Aketch, E. O. (2026). “Continued Fractions: Theory, Convergents and Best Rational Approximations.” July 15, 2026. DOI: 10.2139/ssrn.7151399.

DATA SCIENCE & COMPUTATIONAL SKILLS

- Python: data analysis, numerical computation and scientific programming.
- Python ecosystem: NumPy, pandas, Matplotlib, Jupyter Notebook and Streamlit.
- SPSS: statistical analysis.
- LaTeX: mathematical and scientific document preparation.
- Microsoft Excel, Word, PowerPoint and Access; Google Sheets.

RELEVANT TRAINING & PROFESSIONAL DEVELOPMENT

- Data Science Certificate — WorldQuant University, USA (January–June 2025).
- Deep Learning IndabaX Kenya AI Summit 2026 — KEMRI, Nairobi (21–23 July 2026).
- Sustainability Business Leaders Workshop — Bard University, Center for Environmental Policy (December 2022).
- Certificate in Information Management System — Kenya Education Management Institute (KEMI), March 2026.
- Certificate in Effective Presentations — HP Life Foundation, May 2024.
- Certificate in Effective Leadership — HP Life Foundation, April 2024.

PROFESSIONAL EXPERIENCE

Tutorial Fellow — Jaramogi Oginga Odinga University of Science and Technology | January 2026–Present

- Support students in understanding advanced mathematical concepts introduced in lectures.
- Conduct practical activities and learning experiences that connect mathematical concepts with real-world applications.
- Mark and grade assessments and contribute to fair and effective academic evaluation.

Graduate Teacher & Head of Department (Mathematics) — St. Marks Ndwara Secondary School | January 2020–Present

- Teach Mathematics and Business Studies and coordinate the academic work of the Mathematics department.
- Coordinate internal examinations, assessment and academic activities.
- Provide academic leadership and support collaborative planning among teachers.

Graduate Teacher & Head of Environmental and Climate Change Club — Ramba High School | May 2017–December 2019

- Taught Mathematics while reviewing departmental schemes of work.
- Created and implemented environmentally friendly policies and practices and monitored their effectiveness.
- Promoted awareness of environmental issues among students and staff.

LEADERSHIP & ACADEMIC SERVICE

- Mathematics Exhibition Coordinator — Kenya Science and Engineering Fair, Rarieda Subcounty (January 2024–Present).
- Head of Department (Academics) — St. Marks Ndwara Secondary School (August 2021–Present).
- Head of Department (Mathematics) — St. Marks Ndwara Secondary School (May 2020–Present).
- Head of Subject (Mathematics) — Ramba High School (July 2017–December 2019).

PROFESSIONAL MEMBERSHIPS

- African Institute for Mathematical Sciences (AIMS)
- Mathematics Association of Kenya (MAK)

- Teachers Service Commission (TSC)

LANGUAGES

English — Fluent | Kiswahili — Fluent

REFERENCES

Dr. Tchammou Bagnantissoun Euloge, PhD

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Prof. Benard Okello, PhD

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